

Flow-KTO: Learning Flow-Based Robot Policies from Unary Feedback

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1 Introduction

During real-world robot deployment, a policy produces an action trajectory from the current observation, and executing that trajectory changes the physical scene. Collecting an alternative action under exactly the same conditions would require reproducing object poses, contact states, sensor readings, and timing. Such reconstruction is generally unavailable after execution. The deployed trajectory can nevertheless be inspected afterward and labeled as desirable or undesirable. The resulting deployment feedback contains one executed trajectory and one unary judgment for each encountered context, but no same-context counterfactual. It identifies behavior to encourage or suppress without providing the alternative needed for a direct comparison. Reusing this feedback is attractive precisely because it avoids recollecting counterfactual actions or rewarded candidate groups on real hardware. The challenge is to turn this direction-only deployment feedback into a valid policy update.

Flow-based robot policies add a separate optimization constraint. They generate continuous action trajectories by integrating a learned vector field from noise through multiple steps [1, 2]. Their networks therefore do not emit terminal action likelihoods as direct outputs. Naively differentiating a trajectory-level objective through the full generation process can also yield unstable gradients when adapting such a deployed policy. The problem is consequently defined by both the deployment feedback and the policy interface: can a pretrained flow-based robot policy be improved directly from unary feedback collected during real-world deployment, without collecting additional robot interactions for optimization or learning a reward or value critic?

Existing post-training methods obtain a learning signal by adding comparison or value structure. Direct Preference Optimization requires a preferred and a dispreferred output for the same context [3]; Proximal Policy Optimization and Group Relative Policy Optimization instead form advantages from rewarded current-policy trajectories or groups of candidates [4, 5, 6]. Recent flow-policy methods illustrate critic-based alternatives. RECAP converts deployment outcomes into binarized advantage conditioning through a learned value function, using demonstrations, autonomous rollouts, and human corrections [7]. Q-learning with Adjoint Matching learns a temporal-difference critic from reward-labeled transitions and uses adjoint matching to turn its action gradients into supervision for the flow field [8]. These routes are viable under their respective data interfaces, but they do not define the restricted update studied here, whose only quality signal is a fixed set of individually judged deployment trajectories. They first construct pairwise or group comparisons, collect current-policy experience, or learn critic-derived targets. We instead ask whether unary judgments can reach the flow policy without these intermediaries.

Without a learned intermediary, a unary judgment must be converted into a comparison using quantities available from the policy itself. The first comparison asks how the current policy’s fit to a labeled deployment trajectory has changed relative to the frozen pretrained policy. This reveals how training has moved the recorded behavior, but not how that change stands relative to the current policy’s own outputs. A second, policy-centered comparison must therefore come from trajectories generated by the current policy at the stored deployment contexts. Recorded deployment actions cannot provide this reference because they represent the behavior-data distribution; actions borrowed from other contexts would additionally confound policy change with context–action mismatch. Together, the two comparisons anchor each update to the pretrained policy while interpreting it against the policy being optimized.

We propose Flow-KTO to implement this principle. Flow-KTO first measures the change in conditional flow-matching fit for each labeled deployment trajectory relative to a frozen reference policy. It then samples

trajectories internally from the current policy at stored deployment contexts, evaluates each sample under the current and reference policies, and aggregates those comparisons into a dynamic reference point. This construction keeps every labeled comparison anchored to the pretrained policy while locating it within the current policy’s output distribution. The unary label determines which side of this policy-centered reference should receive utility, and the exact bounded KTO objective converts that direction into a policy update [9]. The internally sampled trajectories are not assigned quality labels and are never executed on the robot; they are used only to estimate the current policy’s reference point. Because flow-matching differences need not share the numerical scale of explicit likelihood ratios, Flow-KTO calibrates the inverse temperature to the scale of these comparisons while keeping class-utility and objective-mixing weights separate.

The paper makes three technical contributions:

- We formulate direct post-training of a pretrained flow-based robot policy from a fixed collection of feedback gathered during real-world robot deployment, with one unary-labeled trajectory per encountered context and no requirement for paired alternatives, rewarded groups, additional robot interaction, or a learned reward or value critic.
- We introduce a policy-centered reference estimator in the native flow-matching training space. It uses internally sampled current-policy trajectories at stored deployment contexts and is distinct from references computed on recorded deployment actions or context-mismatched actions.
- We optimize the resulting label-directed comparison with exact bounded KTO and calibrate its inverse temperature to the scale of the flow-matching comparisons, separate from class-utility and objective-mixing weights.

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